Verasonics®
The leader in Research Ultrasound™

Vantage® *NXT* Research Ultrasound Systems

FastSave Setup and Usage Instructions

Verasonics Inc. | 11335 NE 122nd Way, Ste 100, Kirkland, WA 98034 | www.verasonics.com
©Verasonics Inc. 2001- 2025. Verasonics Confidential.

Table of Contents

1. FastSave Overview *2*

2. Recommended Hardware *2*

3. Hardware Setup *2*

3.1 Linux **2**

3.2 Windows **2**

4. Software and Library Setup *2*

4.1 Linux **2**

4.2 Windows **2**

5. Optional: Benchmark Storage with FIO *3*

5.1 Install FIO **3**

5.1.1 Linux 3

5.1.2 Windows 3

5.2 Run the Test Script **3**

5.2.1 Linux 3

5.2.2 Windows 3

6. fastSave MEX function *4*

7. Example Scripts (Provided) *5*

7.1 Script 1 — 8 MHz Vermon 2D matrix (4:1 mux) **5**

7.2 Script 2 — L22-4vX linear probe **5**

7.3 Running the example scripts **5**

7.3.1 Validation and offline reconstruction 5

8. Troubleshooting and recommendations *6*

9. Appendix A: Storage Setup *7*

9.1 Linux (GUI-driven example) **7**

9.2 Windows **10**

List of Figures

Figure 1: Disk Management GUI *7*

Figure 2: Select a Drive *7*

Figure 3: Create Partition *8*

Figure 4: Format Volume *8*

Figure 5: Mount the Drive *9*

Figure 6: Full Path of Mounted Drive *9*

Figure 7: Drive Letters *10*

Figure 8: Disable Drive Indexing *11*

1. FastSave Overview

The FastSave tool for the Verasonics Vantage system enables high throughput streaming of RF data to disk with minimal interference to the real-time acquisition loop. Review the recommended hardware, hardware, software, and library setup procedures, benchmark storage with FIO (Flexible I/O tester) option, and instructions on how to use the example scripts.

2. Recommended Hardware

- 4 × 1 TB Western Digital WD_BLACK NVMe drives (SN510 series is a good option; SN8100 series is higher performance).
- 1 x Dell ultra-speed quad NVMe card (Manufacturer part 7YJH0 | Dell part 400-BPJP)
  Dell reports compatibility with the following models:
  Precision Workstation 5860 Tower
  Precision Workstation 5860XL Tower
  Precision Workstation 7875 Tower
  Precision Workstation 7875XL Tower
  Precision Workstation 7960 Rack
  Precision Workstation 7960 Tower
  Precision Workstation 7960XL Rack
  Precision Workstation 7960XL Tower

3. Hardware Setup

Ensure each target NVMe volume is formatted and mounted before running FastSave.

See Appendix A: Storage Setup for step-by-step GUI and CLI instructions for both Linux and Windows.

3.1 Linux

1. Format each NVMe as ext4.
2. Mount the drives.
3. Note the full mount paths (e.g., /media/verasonics/WD1) and confirm write permissions.

3.2 Windows

1. Format each NVMe as NTFS.
2. Note the drive letters (e.g., E:).

4. Software and Library Setup

4.1 Linux

- FastSave on Linux uses asynchronous I/O. Ensure that libaio is available on your system. On Debian/Ubuntu this can be installed with:

```
bash

sudo apt update
sudo apt install libaio-dev
```

NOTE: The runtime package may be named `libaio1` while development headers are in `libaio-dev`.

4.2 Windows

- No additional system libraries are required for the Windows binaries provided.

5. Optional: Benchmark Storage with FIO

FIO (Flexible I/O tester) is recommended to measure baseline write performance for each NVMe device before using FastSave. Benchmarks help establish realistic expectations for sustained throughput and guide configuration choices (super-frame size, number of concurrent files, etc.).

5.1 Install FIO

5.1.1 Linux

Run the FIO install on Linux (Debian/Ubuntu):

```
bash

sudo apt install fio
```

5.1.2 Windows

A pre-built binary installer can be downloaded from GitHub (version 3.41 at the time this document was written: https://github.com/axboe/fio/releases/download/fio-3.41/fio-3.41-x64.msi).

5.2 Run the Test Script

5.2.1 Linux

1. Edit `fio_linux_test.sh` or `fio_windows_test.ps1` to set the correct drive paths/letters.
2. The optimal FIO parameters for sequential write benchmarking have been set in the scripts:
3. Run the Linux test script:

```
bash

./fio_linux_test.sh
```

5.2.2 Windows

Or run the Windows PowerShell script (you may need to allow script execution):

```
powershell

# from an elevated PowerShell prompt
powershell -ExecutionPolicy Bypass -File .\\fio_windows_test.ps1
```

The supplied MATLAB script `processFIOresults.m` can be used to parse the CSV output and plot throughput over time.

6. fastSave MEX function

- **Purpose**: Asynchronously write RF super-frames to disk with minimal impact on the Vantage acquisition loop.
- **Inputs**: Configuration variables defined in the MATLAB base workspace (see below), plus int16 RF data buffers passed by the Verasonics sequence.
- **Outputs**: One or more binary files on disk containing int16 RF samples (no header); progress and diagnostic messages printed to the MATLAB console on Linux.
- **Error modes**: insufficient disk space, permission/ownership errors, or I/O errors are reported to the MATLAB console.

IMPORTANT:

- Running the fastSave function from the Matlab commandline with a matrix defined from the commandline will result in an empty file. fastSave will only function properly in the context of the Verasonics External Process.
- The fastSave function has been compiled for Windows and Linux and will be named as such: fastSaveLinux.mexa64 and fastSaveWindows.mexw64
- The base workspace must contain the variables used by the save routine. Example required variable:

```
matlab

fsFileName = 'exp17_';  % base filename; FastSave will append a zero-padded index:
exp17_00001.rf
fsDriveNames =
{'/media/verasonics/WD1','/media/verasonics/WD2','/media/verasonics/WD3','/media/ve
rasonics/WD4'}; % Linux example
fsDriveNames = {'E:\','F:\','G:\','H:\'}; % Windows example
```

- File format: each RF file contains exactly one super-frame of interleaved int16 values in column-major order (MATLAB order). There is no file header; therefore, the user must record the frame dimensions (samples per line, number of lines/channels, and number of subframes per super-frame) for post-processing.
- On Linux, FastSave prints write-speed diagnostics to the console that launched MATLAB; this is useful for debugging and performance evaluation.

7. Example Scripts (Provided)

Two example acquisition scripts are included in this folder. They illustrate typical FastSave configurations and demonstrate how to combine super-framing with real-time reconstruction.

7.1 Script 1 — 8 MHz Vermon 2D matrix (4:1 mux)

- Target probe: 8 MHz Vermon 1024-element 2D matrix with 4:1 multiplexing.
- RF time tags enabled for acquisition timing validation.
- Oversampling: 8 samples/wavelength (for performance testing).
- Acquisition: 5 angled plane waves with spatial compounding.
- Imaging mode: "Light" sequence (trade-off between image quality and frame rate).
- Super-frame: 26 sub-frames collected and DMA'd as a single block to increase DMA efficiency.
- Real-time reconstruction: reconstructs the first sub-frames of each super-frame to provide monitoring at reduced rate (example: ~3.8 fps for the provided configuration).

7.2 Script 2 — L22-4vX linear probe

- Target probe: L22-4vX (Canon ZIF connector).
- RF time tags enabled.
- 7 angled plane waves with spatial compounding.
- Based on the `L22-4vFlashAngles` example with FastSave + superframes modifications.
- Super-frame example: 50 sub-frames per super-frame; live imaging ~18 fps (subject to hardware limits).
- Real-time reconstruction: reconstructs the first sub-frames of each super-frame to provide monitoring at reduced rate (18 fps).

7.3 Running the example scripts

1. Start acquisition using the Verasonics GUI button labeled "Start Acquisition". This will enable the FastSave save event to hand data to the external save process.
2. Press the same button again to stop saving; the acquisition sequence can continue without writing to disk.

7.3.1 Validation and offline reconstruction

- Validate collected data with `validateCollectedData.m`. This script loads the saved RF files, examines embedded time-tags, converts to milliseconds, and plots the period between acquisitions. Deviations from the expected period indicate dropped or delayed acquisitions.
- Offline reconstruction: `process_8MHz_FlashAngles_Light_superframe.m` demonstrates how to reconstruct saved RF data into IQData and ImgDataP files.
- Generate movies from reconstructed frames using `imgdatapReplay.m`.

8. Troubleshooting and recommendations

- To obtain the performance for an NVMe drive a TRIM operation should be applied to the intended drives. This allows the drive to pre-emptively erase data blocks so that new data can be written directly to empty pages without waiting for a separate deletion process. There are several methods for doing this with the simplest simply being to reformat the drives. A more direct manual method is:
  (Windows) Open the drive defragmenation tool and click optimize for each destination drive.
  (Linux) Install the `fstrim` tool and use the command `sudo fstrim -av` for the intended destination drives.
- Using multiple independent NVMe devices and writing separate files to each device (striping at the application level) often yields higher sustained throughput than concatenating devices via RAID0 for this workload. This avoids RAID controller overhead and gives explicit control over file placement.
- Ensure the user has write permission to the target mount points.
- Monitor available disk space during long acquisitions.
- FastSave does not add file headers with metadata, so data dimesions must be recorded by the user independtly to post process the RF Data.
- If sustained write performance is below expectations, run fio and verify your drive is performing as expected.

9. Appendix A: Storage Setup

9.1 Linux (GUI-driven example)

1. Open Disk Management GUI.

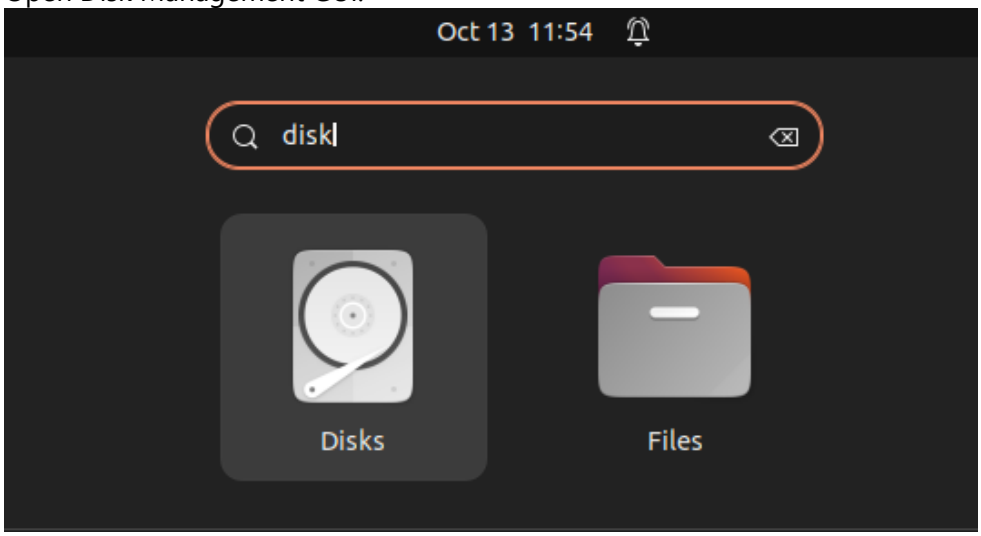

Figure 1: Disk Management GUI

2. Select a drive to partition and format.

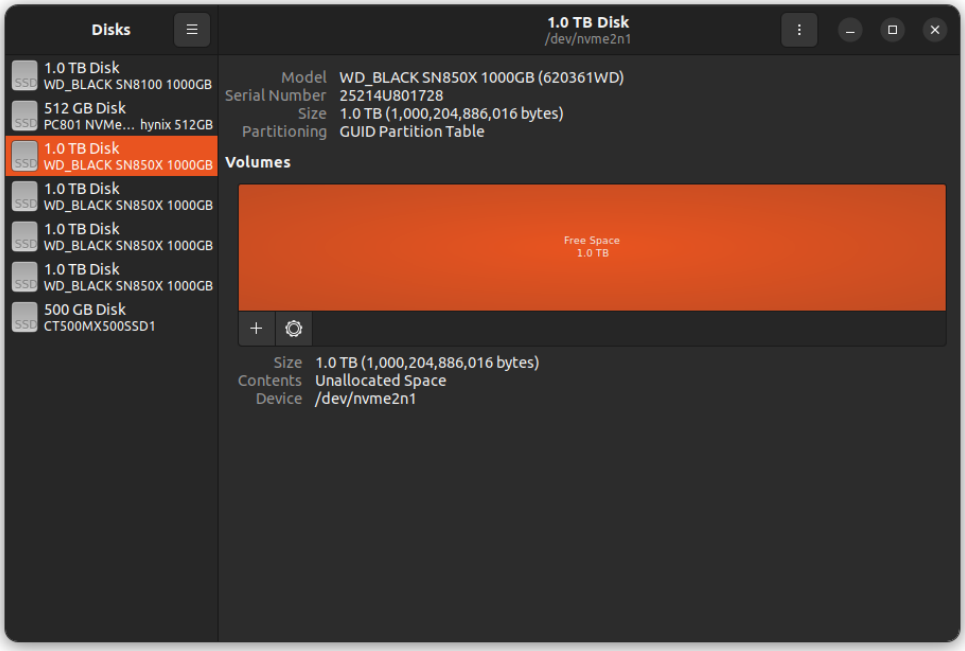

Figure 2: Select a Drive

3. Click the "+" to create a partition.

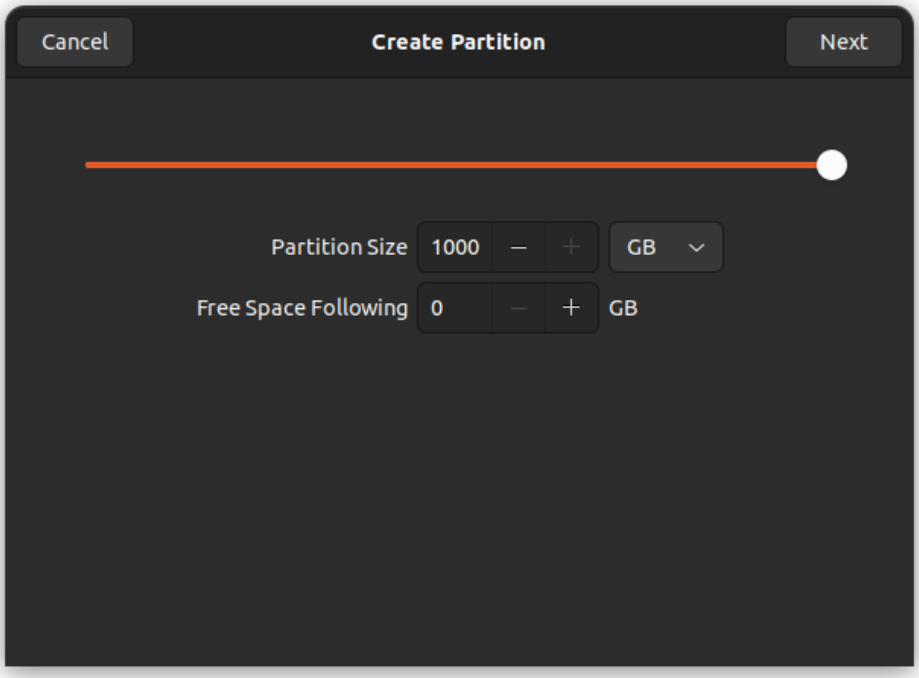

Figure 3: Create Partition

4. In the **Volume Name** field, type the name of the partition. In this example, the first drive is named "WD1". Keep the default value of "Ext4".

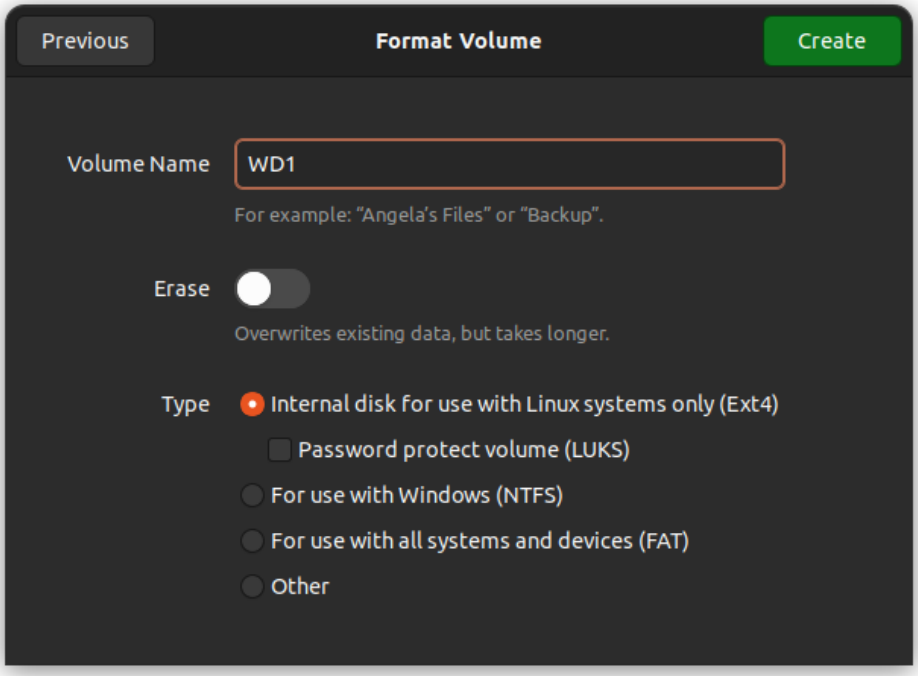

Figure 4: Format Volume

5. Click **Create**. The drive will be formatted but will not be mounted automatically.

6. Click the right-facing triangle to mount the drive.

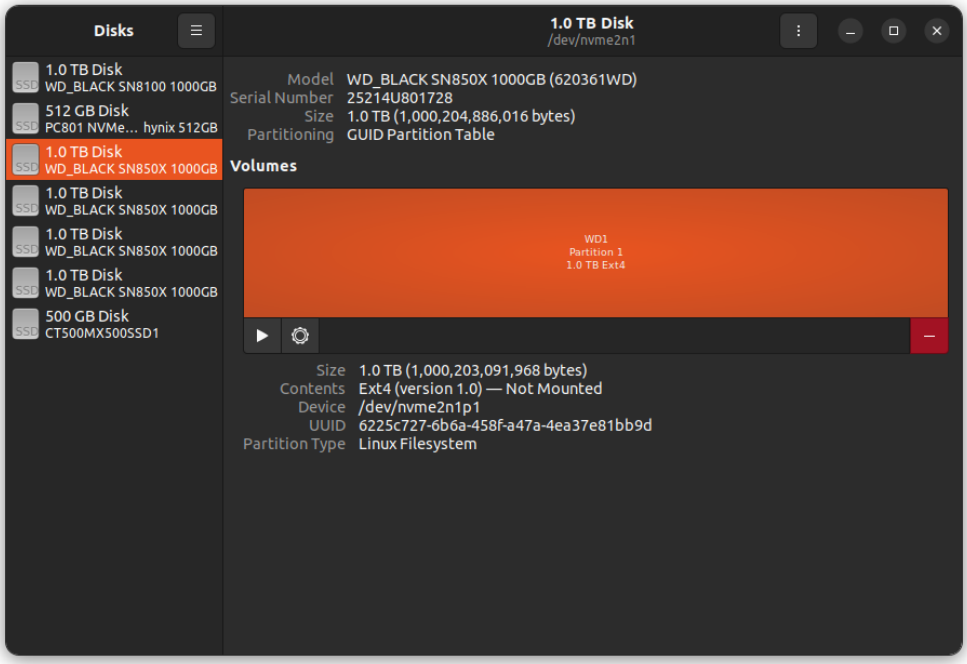

Figure 5: Mount the Drive

7. Make a note of the full path for the mounted drive. This will be referenced in the sequence script.

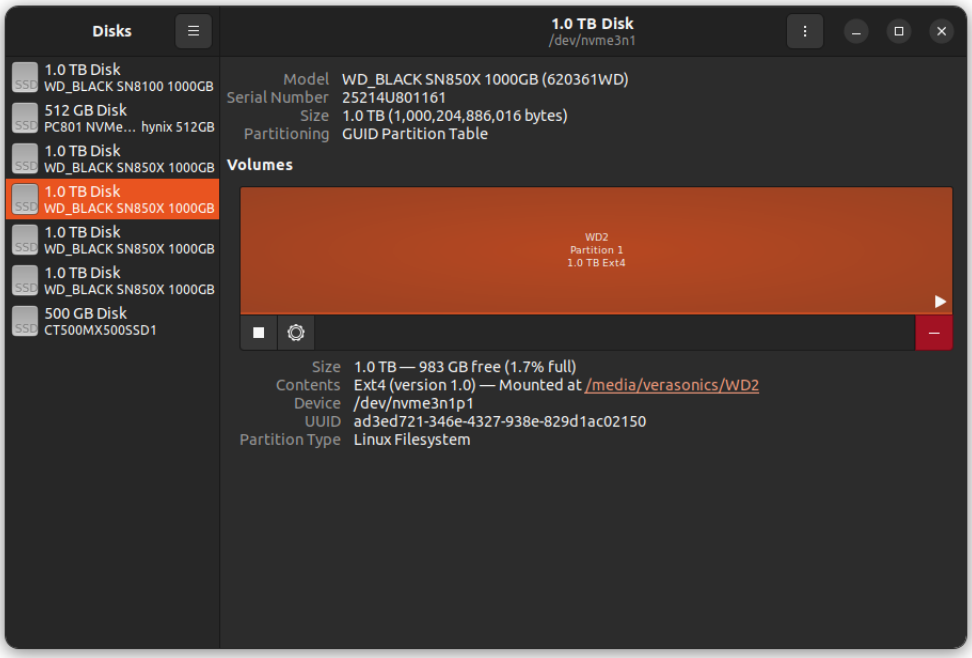

Figure 6: Full Path of Mounted Drive

In this example, the 2nd drive has been mounted as “/media/verasonics/WD2”.

9.2 Windows

1. Format each NVMe volume as NTFS (do not use ReFS or FAT for this application).
2. Make a note of the drive letters (for example: D:, E:, F:, G:) and use them when configuring the sequence and save scripts.

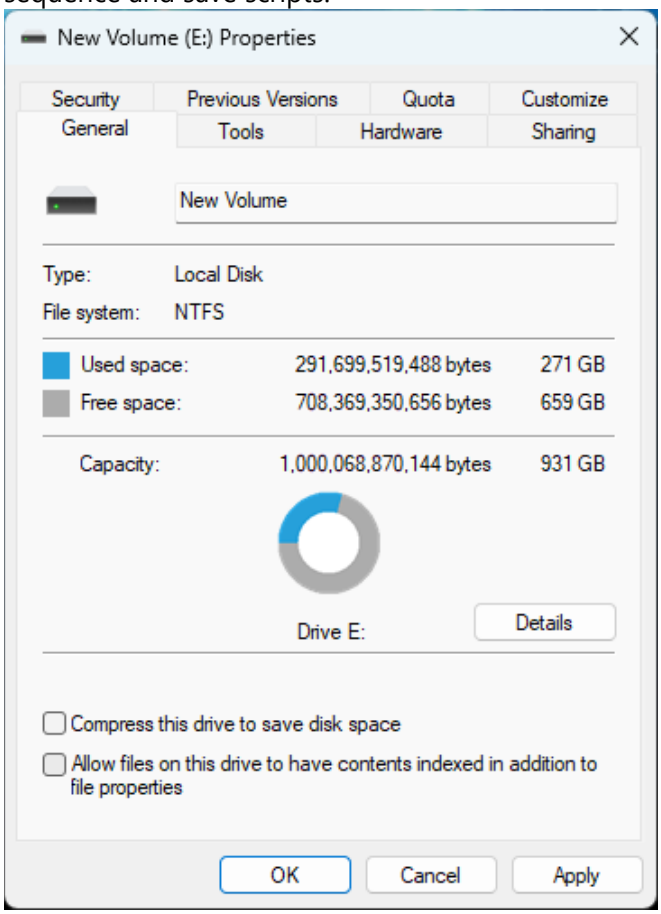

Figure 7: Drive Letters

3. Disable drive indexing for search.

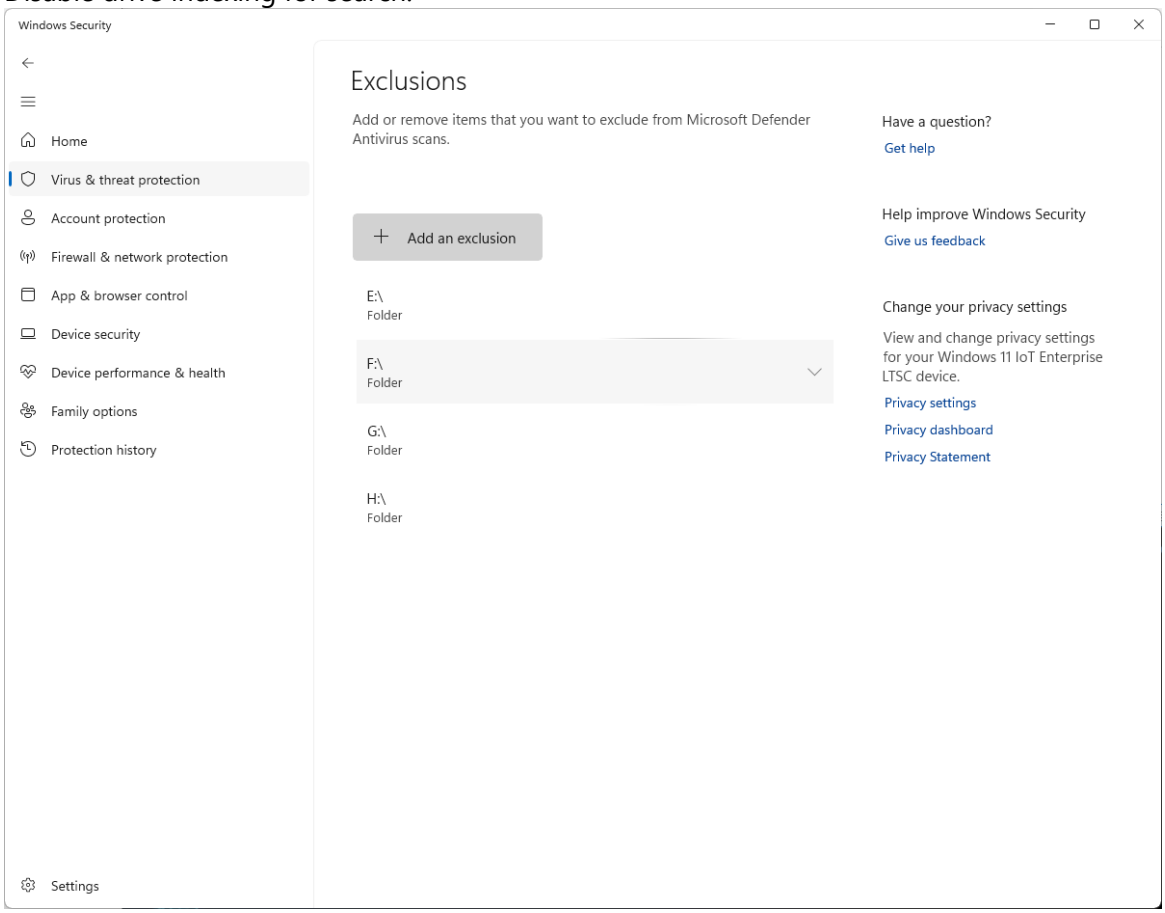

Figure 8: Disable Drive Indexing

4. Exclude the drive from Windows Defender.

This document is intended for use by Verasonics' customers only.
Do not duplicate or distribute without permission.